

JERÓNIMO GARCÍA-MEJÍA

PERSONAL DATA

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RESEARCH INTERESTS: Geometric Group Theory and Geometric Topology.

ACADEMIC POSITIONS

OCT 2025 – PRESENT UNIVERSITY OF WARWICK
Research Fellow
Mathematical Institute
Postdoctoral Mentor: Dr. Ric Wade
DEC 2024 – OCT 2025 UNIVERSITY OF OXFORD
Postdoctoral Research Associate
Mathematical Institute
Postdoctoral Mentor: Dr. Ric Wade

EDUCATION

SEP 2024 KARLSRUHE INSTITUTE OF TECHNOLOGY
PhD in Mathematics
Thesis: *Dehn functions of groups with filiform subgroups*
Advisor: Jun.-Prof. Claudio Llosa Isenrich
AUG 2020 UNIVERSITY OF BONN
M.Sc. in Mathematics
Thesis: *Group actions on hyperbolic spaces*
Advisor: Prof. Ursula Hamenstädt
APR 2018 UNIVERSIDAD NACIONAL AUTÓNOMA DE MÉXICO
B.Sc. in Mathematics
Thesis: *Construction of classifying spaces* (in Spanish)
Advisor: Prof. Carlos Prieto

SCHOLARSHIPS

2018–2020 ROSA LUXEMBURG STIFTUNG (Rosa Luxemburg Foundation)
Scholarship

PUBLICATIONS

- Yu-Chang Chan, Jerónimo García-Mejía, and Matteo Migliorini, *Complete classification of the Dehn functions of Bestvina–Brady groups*.
Geometric and Functional Analysis (2026).
- Jerónimo García-Mejía, Claudio Llosa Isenrich and Gabriel Pallier *On Dehn functions of central products of nilpotent groups*.
Transactions of the American Mathematical Society (2026).
- Jerónimo García-Mejía and Antoine Goldsborough, *Random Dehn function of groups*.
Journal of Topology and Analysis (2026).

OTHER WORK CONTAINING ORIGINAL RESEARCH

- *Dehn functions of groups with filiform subgroups.*
PhD thesis, Karlsruhe Institute of Technology.
- *Group actions on hyperbolic spaces.*
Master thesis, University of Bonn. Unpublished.

INVITED CONFERENCE TALKS

- OCT 2026 *TBC, Groups and Geometry in the South East Meeting*, University College London. (scheduled).
- OCT 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, *Conference: LiT IV Landing in Topology*, Università di Pisa (scheduled).
- SEP 2026 *Distortion and coarse median preserving automorphisms of special groups*, *Conference: Automorphisms groups in Bonn*, Universität Bonn.
- JUN/JUL 2026 *La geometría del problema de la palabra* *Summer School: Escuela de Teoría Geométrica de Grupos*, Instituto de Matemáticas Unidad Oaxaca, México.

INVITED SEMINAR TALKS

- MAY 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, *Geometry and Topology Seminar*, University of Glasgow.
- APR 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, *Oberseminar Geometric Group Theory*, Universität Bonn.
- JAN 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, *Topology Seminar*, University of Oxford.
- JAN 2026 *Complete classification of the Dehn functions of Bestvina–Brady groups*, *Geometry and Topology Seminar*, University of Warwick.
- NOV 2025 *Complete classification of the Dehn functions of Bestvina–Brady groups*, *Algebra Seminar*, University of Bristol.
- MAY 2025 *Dehn functions of Bestvina–Brady groups*, *Geometric Group Theory Seminar*, University of Cambridge.
- JULY 2024 *Dehn functions of nilpotent groups*, *Geometric Group Theory Seminar*, University of Oxford.
- APR 2024 *Dehn functions of nilpotent groups*, *Research seminar Model theory and groups*, Universität Münster.
- MAR 2024 *Funciones de Dehn de grupos nilpotentes*, *Long-distance seminar in Geometric Group Theory in Mexico*.
- JAN 2024 *Dehn functions of central products of nilpotent groups*, *Geometry and Dynamics Seminar*, Université de Lille.
- OCT 2022 *The Geometry of the Word Problem*, *Baby Geometri Seminar*, Università di Pisa - Scuola Normale Superiore
- MAY 2020 *Group actions on hyperbolic spaces*, *Junior Geometry Seminar*, Universität Heidelberg.

CONTRIBUTED TALKS

- MAY 2026 *Lightning Talk*, IMBM Istanbul Topology Days:: A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond, Boğaziçi University, Istanbul, Turkey.
- MAY 2023 *Lightning Talk*, Conference Geometry of subgroups, Thematic program on Geometric Group Theory, Centre de Recherches Mathématiques, Montréal.
- FEB 2023 *Lightning Talk*, Young Geometric Group Theory XI, Münster.
- FEB 2021 *Group actions, hyperbolic spaces and small cancellation*, AG Seminar Metric Geometry, Karlsruhe Institute of Technology, Karlsruhe.

CONFERENCES AND WORKSHOPS ATTENDANCE

MAY 2026	<i>IMBM Istanbul Topology Days: A Conference on Developments in Contact Topology, Mapping Class Groups, and Beyond</i> , Boğaziçi University, Turkey.
MAR 2026	<i>Topological and Homological Methods in Group Theory 2026</i> , Bielefeld University, Germany.
MAR 2026	<i>Journées Trans-Couesnon de théorie des groupes</i> , University of Caen, France.
SEP 2025	<i>Actions on graphs and metric spaces</i> part of the program <i>Operators, Graphs, Groups</i> , Isaac Newton Institute for Mathematical Sciences, United Kingdom.
JUL 2025	<i>Non-positive curvature and applications</i> , Isaac Newton Institute for Mathematical Sciences, United Kingdom.
JUN 2025	<i>The William Rowan Hamilton Geometry and Topology Workshop: Celebrating Martin Bridson's 60th Birthday</i> , The Hamilton Mathematics Institute, Trinity College Dublin, Ireland.
MAY 2025	<i>Groups and Geometry in South England</i> , University of Oxford, United Kingdom.
AUG 2024	<i>Workshop on Profinite Rigidity</i> , Karlsruhe Institute of Technology, Germany.
APR 2024	<i>Young Geometric Group Theory XII</i> , University of Bristol, United Kingdom.
MAR 2024	<i>Topological and Homological Methods in Group Theory 2024</i> , Bielefeld University, Germany.
JUN 2023	<i>Cube complexes and combinatorial group theory</i> part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Canada.
JUN 2023	<i>Groups around 3-manifolds</i> part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Canada.
MAY 2023	<i>Geometry of subgroups</i> part of the Thematic program on Geometric group theory, Centre de Recherches Mathématiques, Canada.
FEB 2023	<i>Young Geometric Group Theory XI</i> , Münster University, Germany.
JUL 2022	<i>Metric Geometry and Geometric Analysis</i> , MSRI-Oxford Summer Graduate School, Oxford University, United Kingdom.
JUN 2022	<i>Hyperbolic groups and their generalisations</i> , Institut Henri Poincaré, France.
MAR 2022	<i>Topological and Homological Methods in Group Theory 2022</i> , Bielefeld University, Germany.
FEB 2022	<i>Workshop on finiteness properties</i> , Karlsruhe Institute of Technology, Germany.

EXTENDED VISITS

OCT 2025 – JAN 2026	UNIVERSITY OF OXFORD, MATHEMATICAL INSTITUTE Academic visitor
SEP 2025	ISAAC NEWTON INSTITUTE Semester program <i>Operators, Graphs, Groups</i>
MAY – JUN 2023	CENTRE DE RECHERCHES MATHÉMATIQUES Thematic program on <i>Geometric group theory</i>

TEACHING EXPERIENCE

My teaching experience as *Teacher's Assistant* covers preparing exercise-sheets, presenting solutions in front of the class, and grading. All of this in Spanish, English, and German.

WINTER 2021/22	<i>Elementare Geometrie (Elementary Geometry)</i> , Bachelor course in German at the Karlsruhe Institute of Technology.
SUMMER 2021	<i>Geometric Group Theory</i> , Master course in English at the Karlsruhe Institute of Technology.
WINTER 2020/21	<i>Proseminar Geometrie: Knotentheorie (Knot theory)</i> , Bachelor Seminar in German at the Karlsruhe Institute of Technology.
WINTER 2017/18	<i>Analytical Geometry I</i> and <i>Topology I</i> , Bachelor courses in Spanish at Universidad Nacional Autónoma de México.

SUPERVISION EXPERIENCE

- JUL – SEP 2026 I supervised a 3rd year student as part of the *Undergraduate Research Support Scheme* at the University of Warwick.
The project was titled *Cops and Robbers in Geometric Group Theory*. It consisted on determining the “strong cop number” for groups.
- JUL – AUG 2026 I supervised a group of three students as part of the *Summer Research programme for female-identifying students* at the University of Warwick.
The project was title *The geometry and combinatorics of geodesics in nilpotent groups*. It consisted on constructing explicit combings for nilpotent groups and abelian-by-abelian groups.

SERVICES

- PRESENT Organiser of the Warwick Geometry and Topology Seminar.
- AUG 2024 I co-organised of the *workshop Profinite Rigidity*, joint with the University of Münster, Regensburg University, and the Karlsruhe Institute of Technology, Germany.
- APR 2022 I co-organised of the *Seminar Group (co)homology* of the “RTG Asymptotic Invariants and Limits of Groups and Spaces” joint with the Karlsruhe Institute of Technology and Heidelberg University, Germany.
- OCT 2021 – SEP 2022 I was a speaker of the RTG “Asymptotic Invariants and Limits of Groups and Spaces”, Germany.